

## 5. Area and Perimeter of Rectangle, Circle, Triangle and Square

```
#include <iostream>
using namespace std;

const float PI = 3.14159;

int main() {
    float length, breadth, radius, base, height, side;

    // Rectangle
    cout << "Enter length and breadth of rectangle: ";
    cin >> length >> breadth;
    cout << "Rectangle Area = " << length * breadth << endl;
    cout << "Rectangle Perimeter = " << 2 * (length + breadth) << endl;

    // Circle
    cout << "\\n\\nEnter radius of circle: ";
    cin >> radius;
    cout << "Circle Area = " << PI * radius * radius << endl;
    cout << "Circle Perimeter = " << 2 * PI * radius << endl;

    // Triangle
    cout << "\\n\\nEnter base and height of triangle: ";
    cin >> base >> height;
    cout << "Triangle Area = " << 0.5 * base * height << endl;

    // Equilateral triangle perimeter
    cout << "Enter all three sides of triangle: ";
    float a, b, c;
    cin >> a >> b >> c;
    cout << "Triangle Perimeter = " << a + b + c << endl;

    // Square
    cout << "\\n\\nEnter side of square: ";
    cin >> side;
    cout << "Square Area = " << side * side << endl;
    cout << "Square Perimeter = " << 4 * side << endl;

    return 0;
}

/* Output:
Enter length and breadth of rectangle: 10 5
Rectangle Area = 50
Rectangle Perimeter = 30

Enter radius of circle: 7
Circle Area = 153.938
Circle Perimeter = 43.9823

Enter base and height of triangle: 10 5
Triangle Area = 25
Enter all three sides of triangle: 3 4 5
Triangle Perimeter = 12

Enter side of square: 4
Square Area = 16
Square Perimeter = 16
*/
```